SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE CODE: CSA1606 COURSE NAME: Data Warehousing and Data Mining

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COURSE OUTCOME COVERED

CO2: Apply suitable Data Pre-processing techniques like Data cleaning, Data integration, Data transformation and data Reduction, for effective data preparation (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 7 **Total Marks: 50**

Annexure A

SENARIO BASED QUESTIONS

Code of Conduct:

I, Sameer Ahmed G (Register No. 192421154), certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Sameer Ahmed G

RUBRICS FOR CALCULATION:

Criteria	Understanding of Scenario	Identification of Key Issues	Application of Concepts	Decision Making	Clarity of Response	Total (50) Marks	Out of 100
Weightage	20%	20%	25%	20%	15%	100%	
Q1							
Q2							
Q3							
Q4							
Q5							

Signature of the Faculty **Total Marks Awarded :**

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Annexure A

1. Smart Traffic Monitoring

Traffic sensors collect city data.

Sensor_ID	Speed (km/h)	Vehicle_Count	Signal_Time	Area_Type	Congestion
T1	40	200	60	Urban	Low
T2	NaN	180	55	Urban	Medium
T3	50	NaN	70	HIGHWAY	Low
T4	200	1000	200	Highway	High
T5	45	210	NaN	Urban	Medium

Tasks
A. Missing value treatment
B. Detect outliers in Speed and Vehicle_Count
C. Standardize Area_Type
D. Normalize Signal_Time
E. Identify mining task

2. Movie Recommendation System

Streaming platform analyzes user behavior.

User ID	Watch Time	Ratings	Reviews	Genre	Preference
U1	10	4	5	Action	High
U2	NaN	3	4	Action	Medium
U3	12	Nan	6	Drama	High
U4	100	10	50	Drama	Low
U5	8	2	Nan	Action	Medium

Tasks
A. Handle missing values
B. Detect outliers
C. Standardize Genre
D. Discretize Ratings
E. Mining task

3. **Fitness Tracking Dataset**

A wearable device collects user data.

User ID	Steps	Calories	Sleep (hrs)	Gender	Fitness_Level
F1	8000	2000	7	Male	Good
F2	NaN	1800	6	Male	Average
F3	9000	NaN	8	Female	Poor
F4	50000	10000	20	Female	Poor
F5	7000	1500	NaN	Female	Average

Tasks

A. Missing value handling
B. Detect outliers
C. Standardize Gender
D. Normalize Steps
E. Mining task

4. Weather Prediction System

Weather stations collect environmental data.

Station ID	Temp (°C)	Humidity	Rainfall	Region	Weather
W1	30	70	100	Coastal	Rainy
W2	NaN	65	80	Coastal	Cloudy
W3	32	NaN	90	Inland	Sunny
W4	80	10	500	Inland	Storm
W5	28	75	NaN	Coastal	Rainy

Tasks

A. Handle missing values
B. Detect outliers
C. Standardize Region
D. Normalize Rainfall
E. Identify mining functionality

5. Employee Performance Analysis

HR collects performance metrics.

Emp_ID	Work_Hours	Projects	Salary (₹K)	Dept	Rating
E1	40	5	30	IT	High
E2	NaN	4	35	IT	Medium
E3	45	NaN	40	HR	High
E4	20	20	200	HR	Low
E5	38	3	NaN	HR	Medium

Tasks

A. Missing value imputation
B. Outlier detection
C. Standardize Dept
D. Discretize Salary
E. Mining task selection

6. Supply Chain Optimization

Company tracks logistics data.

Order_ID	Delivery_Time	Distance	Cost (₹K)	Transport	Status
O1	5	100	10	Truck	On-Time
O2	NaN	120	12	Truck	Delayed
O3	6	NaN	11	SHIP	On-Time
O4	50	1000	200	Ship	Delayed
O5	4	90	NaN	Truck	On-Time

Tasks

A. Missing value treatment
B. Detect outliers
C. Standardize Transport
D. Normalize Distance
E. Mining task

7. Banking Fraud Detection

A bank analyzes transaction data.

Txn_ID	Amount (₹K)	Frequency	Balance (₹K)	Mode	Fraud
T1	10	5	50	Online	No
T2	NaN	8	60	Online	No
T3	20	NaN	55	CARD	Yes
T4	500	50	2000	Card	Yes
T5	12	6	NaN	Online	No

Tasks

A. Data cleaning
B. Detect abnormal transactions
C. Standardize Mode
D. Discretize Amount
E. Mining task selection

ANNEXURE B - DETAILED SOLUTIONS

The following solutions address all five preprocessing tasks for each scenario. For missing numerical values, the median of the available values is used because it is less sensitive to extreme observations than the mean. Outliers are flagged for checking rather than automatically deleted; they may represent valid events or data-entry errors.

Normalization uses min-max scaling: $x' = (x - \text{minimum}) / (\text{maximum} - \text{minimum})$. The resulting values lie between 0 and 1. Category labels are standardized by correcting spelling and capitalization and using one consistent category name.

Preprocessing methods used

Technique	How it is applied in these scenarios
Missing-value treatment	Use the median of available values in the same numerical column; record which values were imputed.
Outlier detection	Flag unusually large / small values using comparison with the rest of the data and domain context; validate before deletion.
Categorical standardization	Trim spaces and unify spelling / capitalization (e.g., HIGHWAY → Highway, SHIP → Ship).
Min-max normalization	$x' = (x - \min) / (\max - \min)$, giving scaled values from 0 to 1.
Discretization	Convert a continuous value into named intervals (e.g., Low, Medium, High); state bin boundaries explicitly.
Mining-task selection	Choose classification when a known label is predicted, regression for a numeric target, clustering for groups, or anomaly detection for unusual records.

Important: These are proposed preparation choices for the small sample provided in the paper. In a real deployment, confirm units, business rules, and sensor/device context before changing or removing a value.

Q1. Smart Traffic Monitoring

The dataset contains missing readings, inconsistent area labels, and extreme speed / vehicle-count values. These must be handled before congestion analysis.

A. Missing value treatment: Speed for T2 is missing, Vehicle_Count for T3 is missing, and Signal_Time for T5 is missing. Impute each using the median of the available values in its own column.

Attribute	Observed values (sorted)	Median used	Imputation
Speed (km/h)	40, 45, 50, 200	47.5	T2 Speed = 47.5 km/h
Vehicle_Count	180, 200, 210, 1000	205	T3 Vehicle_Count = 205
Signal_Time	55, 60, 70, 200	65	T5 Signal_Time = 65

The median method preserves all sensor rows and limits the influence of extreme observations while filling gaps. The imputed value should be marked in the data audit log so it is not mistaken for a directly measured reading.

B. Detect outliers in Speed and Vehicle_Count: T4 has Speed = 200 km/h and Vehicle_Count = 1000, while the remaining speeds are 40–50 km/h and remaining vehicle counts are 180–210. Both T4 values are suspicious extremes and should be flagged for validation against sensor logs. A rule-based check or IQR analysis can be used; with only five records, domain knowledge should be considered alongside statistical cut-offs. Do not delete the record until confirmed erroneous.

C. Standardize Area_Type: Convert inconsistent text values to a common spelling and capitalization: Urban remains “Urban”; “HIGHWAY” and “Highway” become “Highway”. The standardized values are therefore Urban, Urban, Highway, Highway, Urban for T1–T5. This prevents one real category being split into two labels.

D. Normalize Signal_Time: After imputing T5 as 65, the minimum is 55 and maximum is 200. Apply $x' = (x - 55) / (200 - 55) = (x - 55) / 145$.

Sensor	Signal_Time	Calculation	Normalized
T1	60	(60 − 55) / 145	0.0345
T2	55	(55 − 55) / 145	0.0000
T3	70	(70 − 55) / 145	0.1034
T4	200	(200 − 55) / 145	1.0000
T5	65 (imputed)	(65 − 55) / 145	0.0690

E. Identify mining task: The most direct task is classification of Congestion (Low / Medium / High) using speed, vehicle count, signal time, and area type as predictors. Clustering could also group similar traffic conditions if congestion labels were unavailable. The T4 extreme values should be validated first because they could distort the model.

Q2. Movie Recommendation System

The platform uses watch time, ratings, reviews, genre, and preference to discover what users may want to watch.

A. Handle missing values: Use column-wise median imputation for the numerical fields. For Watch Time, observed values 8, 10, 12, 100 give median (10 + 12) / 2 = 11. For Ratings, observed values 2, 3, 4, 10 give median 3.5. For Reviews, observed values 4, 5, 6, 50 give median 5.5.

User / field	Missing field	Median imputation
U2	Watch Time	11
U3	Ratings	3.5
U5	Reviews	5.5

If ratings must be whole numbers, the system should follow the original rating-scale policy (for example, round 3.5 to 4 only if that policy allows it). Retain an imputation flag for modelling and evaluation.

B. Detect outliers: U4 has Watch Time = 100 and Reviews = 50, much higher than the other users (watch time 8–12 and reviews 4–6). Flag these values for review: they may be valid highly engaged-user behaviour, a unit issue, or a data-entry error. Rating = 10 is only an outlier if the platform's allowed scale is below 10; if ratings are on a 1–10 scale, 10 is valid. Confirm the scale before treating it as erroneous.

C. Standardize Genre: Use consistent category labels: "Action" and "Drama". Trim spaces, normalize capitalization, and map variants such as "ACTION", "action", or " Drama " to the standard labels. For algorithms that need numeric input, one-hot encode Genre (Genre_Action and Genre_Drama) rather than assigning arbitrary numeric ranks.

D. Discretize Ratings: Assuming a 1–10 rating scale, convert numeric ratings into ordered bins: Low = 1–3, Medium = 4–6, High = 7–10. Therefore U1 (4) is Medium, U2 (3) is Low, U4 (10) is High, and U5 (2) is Low. U3's missing rating should first be imputed (median 3.5); according to the chosen bin boundaries, 3.5 falls just above Low and can be assigned Medium. State the rating-scale assumption clearly.

E. Mining task: Use a recommendation task: predict items a user may like from ratings, watch time, genre, and past behaviour. Collaborative filtering finds users with similar behaviour; content-based filtering recommends movies with genres or attributes similar to those a user liked. Clustering users into preference groups can support targeted recommendations.

Q3. Fitness Tracking Dataset

The wearable-device data has missing step, calorie, and sleep measurements plus extreme activity values.

A. Missing value handling: Impute each missing numerical value with the median of its observed column: Steps = median(7,000, 8,000, 9,000, 50,000) = 8,500; Calories = median(1,500, 1,800, 2,000, 10,000) = 1,900; Sleep = median(6, 7, 8, 20) = 7.5 hours.

Field	Missing record	Median imputation
Steps	F2	8,500 steps
Calories	F3	1,900 calories
Sleep (hrs)	F5	7.5 hours

B. Detect outliers: F4 reports 50,000 steps, 10,000 calories, and 20 hours of sleep, all far above the other records. Flag the three observations for device/log validation. Some high-activity days may be genuine, but the sleep value may indicate a sensor error or unusual recording window. Keep the original values and document any correction.

C. Standardize Gender: Normalize category spelling and capitalization to “Male” and “Female”. The values in this dataset already follow that format, so no value changes are needed. If a model needs numerical data, use a consistent categorical encoding (for example, one-hot encoding) rather than treating gender as a measured quantity.

D. Normalize Steps: Use min-max normalization after imputation. The minimum is 7,000 and maximum is 50,000; therefore $x' = (\text{Steps} - 7{,}000) / 43{,}000$.

User	Steps used	Normalized steps
F1	8,000	0.0233
F2	8,500 (imputed)	0.0349
F3	9,000	0.0465
F4	50,000	1.0000
F5	7,000	0.0000

Normalization places the step values on a common 0–1 scale, which helps algorithms that are sensitive to feature magnitude. Since the extreme value sets the maximum, it compresses the other normalized values; robust scaling may be considered if that becomes a modelling issue.

E. Mining task: The data can support clustering users into activity/fitness profiles, regression to estimate calories from steps and sleep, or classification of Fitness_Level (Good / Average / Poor). Outlier detection is also useful for identifying abnormal device readings.

Q4. Weather Prediction System

The weather data combines numerical measurements with categorical region and weather labels.

A. Handle missing values: Use medians: Temperature = median(28, 30, 32, 80) = 31°C; Humidity = median(10, 65, 70, 75) = 67.5; Rainfall = median(80, 90, 100, 500) = 95. These values fill W2 Temperature, W3 Humidity, and W5 Rainfall respectively.

Field	Record	Values used	Imputed value
Temperature	W2	28, 30, 32, 80	31°C
Humidity	W3	10, 65, 70, 75	67.5
Rainfall	W5	80, 90, 100, 500	95

B. Detect outliers: W4 has Temperature = 80°C, Humidity = 10, and Rainfall = 500. These values are extreme compared with the other stations (temperature 28–32°C, humidity 65–75, rainfall 80–100). Flag them to check units, sensor calibration, and local conditions. Extreme weather can be real, so validation should precede deletion.

C. Standardize Region: Use the two consistent region labels “Coastal” and “Inland”. Trim any leading/trailing spaces and standardize capitalization so values are grouped correctly in reports and models.

D. Normalize Rainfall: After imputing W5 rainfall as 95, minimum rainfall = 80 and maximum = 500. Use $x' = (\text{Rainfall} - 80) / (500 - 80) = (\text{Rainfall} - 80) / 420$.

Station	Rainfall used	Normalized rainfall
W1	100	0.0476
W2	80	0.0000
W3	90	0.0238
W4	500	1.0000
W5	95 (imputed)	0.0357

E. Identify mining functionality: A suitable task is weather classification: predict Weather (Rainy / Cloudy / Sunny / Storm) from temperature, humidity, rainfall, and region. With time-stamped data, forecasting can predict future measurements or conditions. Outlier detection can identify sensor faults or unusual weather events.

Q5. Employee Performance Analysis

The HR dataset contains missing work hours, project counts, and salaries; the task is to prepare the data for performance analysis.

A. Missing value imputation: Median values are: Work_Hours = median(20, 38, 40, 45) = 39; Projects = median(3, 4, 5, 20) = 4.5; Salary = median(30, 35, 40, 200) = 37.5 (in ₹K). Thus E2 work hours = 39, E3 projects = 4.5, and E5 salary = ₹37.5K.

Field	Record	Imputed value
Work_Hours	E2	39
Projects	E3	4.5
Salary (₹K)	E5	37.5

B. Outlier detection: E4 has 20 projects and salary ₹200K, while the other project counts are 3–5 and salaries are ₹30K–₹40K. These are clear high-end outlier candidates and should be checked. Work_Hours = 20 is lower than 38–45 but may reflect part-time work or leave, so it should be flagged for context rather than automatically treated as an error.

C. Standardize Dept: Use the consistent department labels “IT” and “HR”. Remove extra spaces and use the same capitalization throughout. E1–E2 remain IT; E3–E5 remain HR. Consistent labels prevent duplicate categories during grouping and modelling.

D. Discretize Salary: For a simple illustrative grouping, use Low = salary below ₹35K, Medium = ₹35K–₹50K, and High = above ₹50K. This yields E1 ₹30K = Low; E2 ₹35K = Medium; E3 ₹40K = Medium; E4 ₹200K = High; E5 imputed ₹37.5K = Medium. The thresholds should be adjusted to the organization’s pay bands before operational use.

E. Mining task selection: Classification can predict the categorical Rating (High / Medium / Low) from work hours, project count, salary, and department. Clustering can find employee groups with similar work patterns. Since performance ratings may influence employment decisions, the model should be validated for fairness and should support, not replace, human review.

Q6. Supply Chain Optimization

The logistics dataset records delivery time, distance, cost, transport type, and delivery status. Cleaning helps identify operational delays and cost drivers.

A. Missing value treatment: Use medians: Delivery_Time = median(4, 5, 6, 50) = 5.5; Distance = median(90, 100, 120, 1000) = 110; Cost = median(10, 11, 12, 200) = 11.5 (₹K). Impute O2 delivery time = 5.5, O3 distance = 110, and O5 cost = ₹11.5K.

Field	Missing record	Median imputation
Delivery_Time	O2	5.5
Distance	O3	110
Cost (₹K)	O5	11.5

B. Detect outliers: O4 has Delivery_Time = 50, Distance = 1000, and Cost = ₹200K, much higher than other records. These are strong outlier candidates; check if they represent an unusually long route, exceptional shipment, wrong units, or data-entry error. Do not remove a genuine delayed delivery because it may be important for improving logistics.

C. Standardize Transport: Standardize labels to “Truck” and “Ship”. The value “SHIP” is changed to “Ship”; “Truck” remains “Truck”. This merges spelling/capitalization variants into the same category.

D. Normalize Distance: After imputing O3 distance as 110, minimum distance = 90 and maximum = 1000. Use $x' = (\text{Distance} - 90) / 910$.

Order	Distance used	Calculation	Normalized
O1	100	10 / 910	0.0110
O2	120	30 / 910	0.0330
O3	110 (imputed)	20 / 910	0.0220
O4	1000	910 / 910	1.0000
O5	90	0 / 910	0.0000

E. Mining task: Classification can predict Status (On-Time / Delayed) from delivery time, distance, cost, and transport type. Regression can estimate delivery time or cost, while clustering can group routes with similar logistics profiles. O4 should remain in the dataset if it is a genuine exception.

Q7. Banking Fraud Detection

The bank data includes transaction amount, frequency, balance, payment mode, and a fraud label. Correct preparation is important because suspicious transactions are rare and costly to miss.

A. Data cleaning: Use median imputation for missing numeric values: Amount = median(10, 12, 20, 500) = 16 (₹K); Frequency = median(5, 6, 8, 50) = 7; Balance = median(50, 55, 60, 2000) = 57.5 (₹K). This fills T2 Amount = ₹16K, T3 Frequency = 7, and T5 Balance = ₹57.5K. Also check duplicates, data types, invalid transaction identifiers, and the allowed Fraud values Yes / No.

Field	Record	Imputed value
Amount (₹K)	T2	16
Frequency	T3	7
Balance (₹K)	T5	57.5

B. Detect abnormal transactions: T4 is highly unusual: Amount ₹500K, Frequency 50, and Balance ₹2,000K are far above the other listed records. Flag T4 for verification, but do not assume it is fraud based on magnitude alone. T3 is labelled Fraud = Yes despite a lower amount, showing that abnormality should be assessed across several features and against the confirmed fraud label.

C. Standardize Mode: Convert “CARD” and “Card” to the single label “Card”; keep “Online” unchanged. Standardize case, trim spaces, and validate Mode against allowed categories. This prevents separate counts for what is actually the same payment mode.

D. Discretize Amount: Using the dataset units (₹K), one straightforward set of bins is Low = 0–20, Medium = greater than 20 to 100, and High = greater than 100. After imputation: T1 = 10 Low; T2 = 16 Low; T3 = 20 Low; T4 = 500 High; T5 = 12 Low. The boundaries are illustrative and should be set using the bank’s transaction policy and historical distribution.

Transaction	Amount (₹K)	Bin
T1	10	Low
T2	16 (imputed)	Low
T3	20	Low
T4	500	High
T5	12	Low

E. Mining task selection: The primary task is supervised classification to predict Fraud (Yes / No) using transaction amount, frequency, balance, mode, and other permitted transaction features. Anomaly detection can flag unusual behaviour for investigation. Evaluate recall, precision, F1-score, and the confusion matrix rather than accuracy alone, because missing fraudulent transactions can be more costly than reviewing a legitimate transaction. Txn_ID should be retained for tracking, not used as a predictive feature.